FoodAI: Know What You're Eating

Comprehensive System Architecture, Technical Specification & Repository File Documentation

System Release	v2.4.0 (Production Release)	**Document Date**	November 2026
Target Platforms	Web SPA & Android Native (Capacitor)	**Android Package ID**	`com.foodai.app`
Firebase Project	food-ai-433b7 (#59556325769)	**Core AI Models**	Gemini 2.0 Flash & Gemini Vision
Automated Tests	100% Pass (91/91 Unit & Integration)	**Repository Branch**	main (Synchronized with origin)
Frontend Framework	React 18 + TypeScript + Vite + Tailwind	**Backend Frameworks**	Django 4.2 REST & Node/Express API

Mission Overview: *FoodAI transforms static, deceptive packaged food labels and ambiguous meals into real-time, actionable pre-eating intelligence. By combining high-frequency camera computer vision, Indian FSSAI Front-of-Pack thresholds, multimodal Google Gemini AI, and condition-specific clinical health rules, FoodAI protects users from acute metabolic damage, hidden sugar spikes, and chronic inflammatory additive exposure prior to consumption.*

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3	**Frontend & Native Android Bridge**	React 18 SPA, Capacitor 6 shell, 60ms auto-detection camera scanner, Web Audio API chime.
4	**AI & Clinical Nutrition Engines**	Gemini Vision Plate AI, FSSAI FOPNL thresholds, NOVA groups 1-4, INS additive matrix.
5	**Personalized Medical Alerts & Damage Control**	8 medical presets (Diabetes, BP, CVD, etc.), pre-eating buffers, 24-hr WHO tracker.
6	**Backend Microservices & API Architecture**	Django REST Framework, Node/Express AI proxy, Firebase Admin SDK, MIME file safety.
7	**Security, Identity & Android Cryptography**	Credential Manager Google Sign-In, Keystore SHA-1/SHA-256 keys, Firestore security rules.
8	**Exhaustive Repository File Map (Frontend)**	Detailed file documentation for all 19 React pages, 5 services, and Android project files.
9	**Exhaustive Repository File Map (Backend & Data)**	Detailed documentation for Django apps, Node routes, configs, and REST API endpoints.
10	**Build, Testing & Deployment Pipelines**	Vitest (91/91 passing), Pytest, Gradle APK build, ADB install, Vercel & Render hosting.
11	**Maintenance, Monitoring & Future Roadmap**	In-app GitHub release updates, structured logging, CGM Bluetooth sync, wearable roadmap.

1. Executive Summary & Clinical Mission

Modern consumer packaged food represents a public health crisis. Ultra-processed foods (UPFs) engineered for hyper-palatability rely on chemical additives, hidden refined starches (Maltodextrin, High-Fructose Corn Syrup), and oxidized industrial seed oils. Front-of-package marketing claims—such as 'Natural', 'Whole Grain', or 'Zero Added Sugar'—frequently mislead consumers while masking extreme levels of free sugars, high sodium, and metabolic disrupters. For patients with chronic conditions like Type 2 Diabetes, Hypertension, Cardiovascular Disease, and Non-Alcoholic Fatty Liver Disease (NAFLD), consuming these products leads to immediate physiological harm.

FoodAI ('Know What You're Eating') inverts this dynamic through instantaneous pre-eating intelligence. Rather than recording food logs retroactively, FoodAI intercepts food choices at the point of decision—in supermarkets, kitchens, and restaurants—evaluating the product against official clinical standards and the user's personal health profile.

Strategic Value Pillars

Pillar	Clinical Objective	Technical Architecture
1. Pre-Eating Medical Alerts	Prevent ingestion of contraindicated ingredients.	Algorithmic cross-referencing against 8 medical condition profiles (Diabetes, Hypertension, CVD, Celiac, etc.).
2. FSSAI FOPNL Warnings	Flag high sugar, saturated fat, and sodium products.	Automated weight-percentage thresholding conforming to Indian Food Safety & Standards Authority front-of-pack regulations.
3. Plate AI Vision	Analyze unpacked, home-cooked, and restaurant meals.	Multimodal computer vision pipeline via Google Gemini 2.0 Flash Vision for multi-item segmentation and portion weight estimation.
4. Tactical Damage Control	Mitigate metabolic impact if a treat is consumed.	Physiological buffering protocols: pre-meal soluble fiber (chia/almonds), sodium flushing hydration, and GLUT-4 walking.
5. 24-Hr WHO Safe Tracker	Prevent chronic cumulative additive & sugar toxicity.	Real-time cumulative tracking against WHO ceilings (Free Sugar 25g, Salt 5g, Saturated Fat 20g, Additive load).

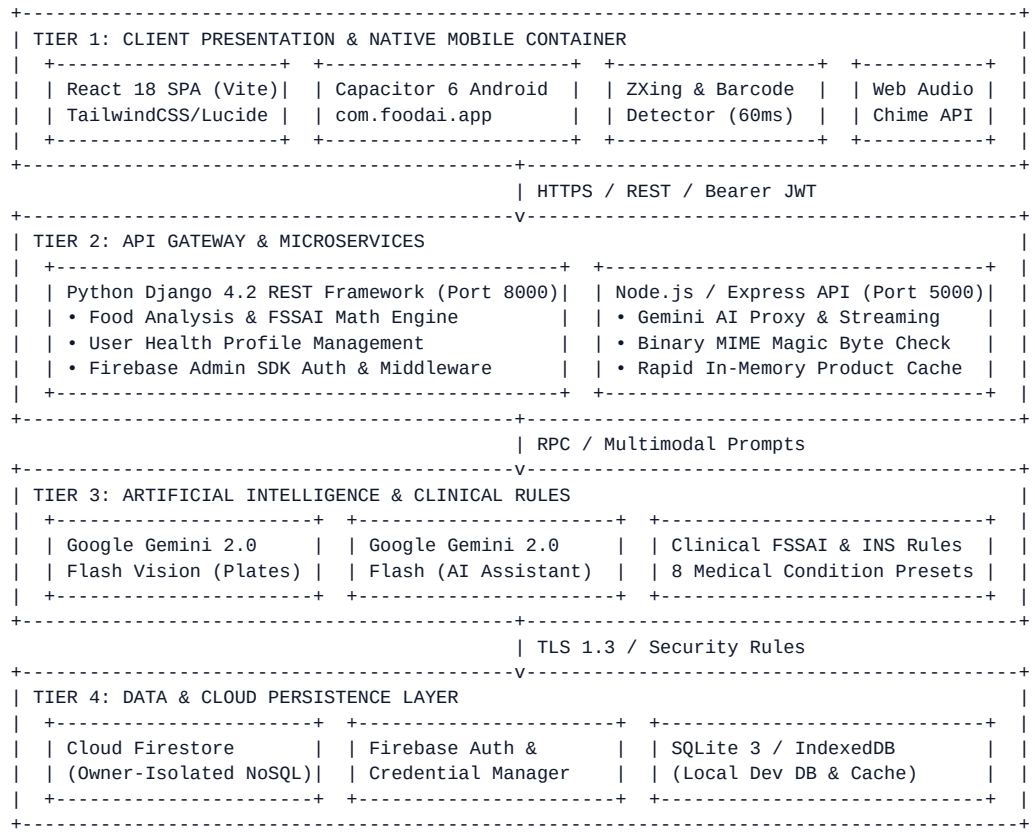
Clinical Personas & User Scenarios

Target Persona	Clinical Need	FoodAI Protective Mechanism
Diabetic & Pre-Diabetic	Avoid postprandial glucose surges and insulin spikes.	Instantly detects maltodextrin, liquid sugars, refined flour (Maida); recommends pre-meal fiber buffering.
Hypertensive Patients	Strict sodium restriction (<1.5% wt / <600mg per 100g).	Flags hidden sodium in condiments, baked goods, and savory snacks; prompts immediate hydration counter-protocols.
Health-Conscious Parents	Shield children from hyperactivity-inducing food dyes.	Child Safe Mode scans for synthetic azo dyes (E102, E110, E129) and high-dose synthetic caffeine.
Fitness & Weight Loss	Precise macro tracking without tedious manual food entry.	Plate AI calculates calories, protein, carbs, and fat from meal snaps with an interactive gram portion slider.

2. High-Level 4-Tier System Architecture

FoodAI is architected as an enterprise-grade, 4-tier hybrid cloud application. The system separates offline-capable client rendering from clinical computational engines and secure cloud storage.

System Architecture Topology Diagram



Tier Responsibilities & Protocols

Architecture Tier	Component	Protocol / Format	Functional Responsibilities
Tier 1: Client	React 18 + Capacitor 6	`Native Webview / HTTPS`	Renders responsive UI. Operates hardware camera, 60ms auto-scanner, tap-to-focus, zoom, torch, audio chime, and Credential Manager.
Tier 2: Backend	Django REST & Express	`REST JSON / Bearer Auth`	Django handles clinical health profiles, FSSAI rules, and catalog queries. Express proxies AI calls with strict MIME-type binary checks.
Tier 3: AI Engine	Gemini 2.0 & Vision	`HTTPS Google GenAI SDK`	Segments compound meal photos, estimates portion weights, calculates calorie/macro totals, and generates conversational clinical advice.
Tier 4: Storage	Firestore & SQLite	`Firestore Rules / SQL`	Cloud Firestore synchronizes user scan history and daily logs with owner-isolation rules. IndexedDB provides zero-latency offline cache.

3. Frontend & Native Android Bridge Architecture

The client application is built with **React 18** and **TypeScript**, bundled via **Vite** into an optimized single-page application. Capacitor 6 packages this bundle inside an Android native container (com.foodai.app), bridging web technologies with hardware Android camera sensors, haptics, and native Google authentication.

Modern Viewfinder & Vision Scanner Subsystem (`src/pages/Scan.tsx`)

- **Ultra-Fast 60ms Auto-Detection Loop:** Employs dual-engine detection. If the native Android browser supports BarcodeDetector with getSupportedFormats(), it utilizes hardware-accelerated C++ decoding. Otherwise, it transparently falls back to @zxing/library continuous multi-format reader.
- **Animated Laser Scanline:** A CSS-accelerated green vertical sweeping laser beam (.laser-line) pulsing continuously between 8% and 92% of the viewfinder height, providing clear feedback of optical scanning.
- **Tap-to-Focus Ring:** Tapping the viewfinder calculates touch coordinates, renders a pulsing target ring, and applies hardware camera constraints via MediaStreamTrack.applyConstraints({ advanced: [{ focusMode: 'continuous' }] }).
- **1x / 2x Optical Zoom Controls:** Interactive pill buttons query camera capabilities (capabilities.zoom) and dynamically toggle between standard (1x) and telephoto/digital crop (2x) zoom.
- **Dark-Environment Auto-Torch:** Analyzes real-time frame pixel luminance; if lighting falls below safety thresholds, it activates or prompts the device flashlight via torch: true.
- **Zero-Dependency Web Audio API Chime:** A synthesized 880Hz → 1760Hz rising frequency tone generated in real time using browser AudioContext oscillators without external media assets.
- **Haptic Confirmation:** Fires navigator.vibrate([40, 30, 80]) upon successful barcode resolution for tactile physical verification.
- **All-in-One Multi-Mode Vision:** Instant segmented tabs: Barcode Scanner, Batch Cart Scanner, In-Viewfinder OCR Text Snapping, and Plate AI.

Android Native Shell Configuration (`frontend/android`)

Resource File	Filesystem Location	Architectural Purpose & Implementation
MainActivity.java	`android/app/src/main/java/com/foodai/app/`	Extends BridgeActivity. Bootstraps Capacitor plugins, initializes the web runtime, and handles Android lifecycle events.
AndroidManifest.xml	`android/app/src/main/`	Declares hardware permissions: CAMERA, INTERNET, CAMERA.AUTOFOCUS, cleartext traffic policy, and FileProvider paths.
google-services.json	`android/app/`	Official Firebase configuration binding package com.foodai.app to Firebase Project #59556325769 and defining OAuth client IDs.
capacitor.config.ts	`frontend/capacitor.config.ts`	Defines appId, webDir (dist), Android scheme (https), and GoogleSignIn scopes (profile, email).
Adaptive App Icons	`android/app/src/main/res/mipmap-*/`	Full adaptive icon suite (mdpi to xxxhdpi) featuring the official FoodAI leaf-shield logo and background layers.

4. Artificial Intelligence & Clinical Nutrition Engines

FoodAI combines algorithmic deterministic clinical rules with generative multimodal AI. This hybrid architecture guarantees mathematical safety compliance while enabling flexible natural vision recognition.

1. Plate AI Vision System (`PlateAnalyzer.tsx` & `geminiVision.ts`)

When analyzing non-packaged, cooked, or restaurant meals, users capture a plate photo. The image is streamed to Google Gemini 2.0 Flash Vision with a rigorous clinical prompt specification:

- **Multi-Item Segmentation:** Deconstructs compound meals into individual food items (e.g., Paneer Butter Masala, Garlic Naan, Basmati Rice, Salad).
- **Volumetric & Portion Estimation:** Estimates realistic weight in grams (g) based on visual plate geometry and standard culinary density constants.
- **Macronutrient Computation:** Computes exact calories (kcal), carbohydrates (g), proteins (g), total fat (g), and dietary fiber (g).
- **Interactive Portion Slider:** Client-side recalculation allows users to fine-tune portion grammage with instant real-time macro re-scaling.
- **Clinical Feedback & Glycemic Impact:** Delivers customized dietary observations, glycemic index predictions, and nutritional advice.

2. Indian FSSAI Front-of-Pack Nutritional Warning Badges (FOPNL)

FoodAI enforces official thresholds set by the Food Safety and Standards Authority of India (FSSAI):

FOPNL Warning Badge	Official FSSAI Threshold	Clinical Risk Justification
■ **HIGH IN SUGAR**	≥ 10.0% by weight (≥ 10g per 100g product)	Induces steep glucose surges, triggers reactive hypoglycemia, promotes hepatic steatosis, and accelerates diabetes.
■ **HIGH IN SATURATED FAT**	≥ 4.0% by weight (≥ 4g per 100g product)	Elevates circulating LDL cholesterol and ApoB particle concentration, contributing directly to atherosclerotic plaque.
■ **HIGH IN SALT / SODIUM**	≥ 1.5% by weight (≥ 600mg sodium per 100g)	Causes extracellular volume expansion, arterial stiffness, and acute blood pressure spikes in hypertensive patients.

3. NOVA Classification & INS Chemical Additive Matrix

Classification Tier	Food Categories & Description	System Processing Guidance
NOVA Group 1: Unprocessed	Whole fruits, vegetables, grains, legumes, eggs, fresh milk, plain meat.	Optimal nutritional baseline. Zero chemical additive load. Safe for daily intake.
NOVA Group 2: Culinary	Oils, butter, sugar, salt extracted from nature for home cooking.	Used for preparation. Monitored for cumulative fat and sodium limits.
NOVA Group 3: Processed	Canned vegetables, artisanal breads, cheeses, cured meats.	Moderate processing. Cross-referenced against salt and preservative thresholds.
NOVA Group 4: Ultra-Processed	Packaged sodas, chips, biscuits, instant noodles, reconstituted meats.	High risk. Flags industrial emulsifiers, artificial flavors, and synthetic sweeteners.

5. Personalized Pre-Eating Health Warnings & Damage Control

FoodAI continuously cross-references product ingredients and macro percentages against the user's active health profile. When an adverse ingredient is identified, FoodAI alerts the user and delivers evidence-based 'Damage Control' mitigation protocols.

Medical Profile	Algorithmic Contraindication Flags	Tactical Damage Control Protocol
■ **Diabetes & Pre-Diabetes**	Free sugars, Maida (refined flour), maltodextrin, high GI starches.	Pre-load 10g soluble fiber (chia/flaxseed) or raw almonds 5 mins before eating to blunt the postprandial glucose spike by up to 35%.
❤■ **Hypertension & High BP**	Sodium >600mg/100g, monosodium glutamate (E621), baking soda.	Drink 400ml water immediately to stimulate renal natriuresis (sodium flushing); consume potassium-rich foods (coconut water, banana).
■ **CVD & Cholesterol**	Palm oil, hydrogenated fats, trans-fats, saturated fat >4%.	Pair with natural plant sterols or omega-3 fatty acids; avoid combining with high glycemic carbs that accelerate LDL oxidation.
■ **Child Safe Mode**	Synthetic azo dyes (E102 Tartrazine, E110, E129), caffeine, aspartame.	Strict consumption block: synthetic food colors are clinically associated with ADHD and hyperactivity in children under age 12.
■ **Celiac & Gluten Sensitivity**	Wheat, barley, rye, malt extract, hydrolyzed wheat protein.	Zero-tolerance warning flag; highlights certified gluten-free alternative products in catalog.
■ **Lactose Intolerance**	Milk solids, whey powder, casein, lactose derivatives.	Recommends lactase enzyme supplementation prior to ingestion; suggests plant-based substitutes (oat/almond/soy).
■ **Kidney & Renal Care**	Potassium sorbate (E202), sodium phosphates, high mineral salts.	Alerts on inorganic phosphate and potassium preservatives that are 90-100% bioabsorbed by compromised kidneys.
■ **Fatty Liver (NAFLD)**	High Fructose Corn Syrup (HFCS), liquid sugars, invert syrup.	Take a brisk 15-minute walk post-meal to activate skeletal muscle GLUT-4 glucose transporters and bypass hepatic de novo lipogenesis.

24-Hour WHO Safe Ceilings & Grocery Basket Screener

Cumulative 24-Hour Tracker (`/tracker`): Monitors free sugar against the WHO daily ceiling of 25g, dietary salt against the 5g ceiling, saturated fat against 20g, and counts cumulative synthetic chemical additives.
Pre-Checkout Grocery Basket Screener (`/cart`): Evaluates shopping baskets prior to supermarket checkout, assigns an overall Basket Health Grade (A to D), isolates the single worst ultra-processed product, and recommends clean-label swaps.

6. Backend Microservices & API Architecture

FoodAI uses a polyglot microservice backend topology. This divides compute workloads into specialized runtimes: a Python Django service for clinical evaluation and data modeling, and a Node.js/Express service for high-performance AI streaming.

A. Python Django REST Framework Service (`foodai_backend`)

- **food_analysis App:** Core engine containing classifier.py (additive detection & ingredient categorizer), health_evaluator.py (medical condition cross-referencing and damage mitigation generation), and nutrition.py (weight percentage calculations and FSSAI badge determination).
- **users App:** Manages user profiles, medical condition preferences, age, and body weight.
- **scans App:** Persists historical barcode scans, OCR extractions, and user audit trails.
- **Firebase Auth Middleware:** Custom FirebaseAuthenticationBackend extracts Firebase JWT Bearer tokens, validates signatures against Google's public key certificates, and automatically attaches or creates local Django user instances.
- **Dataset Management Commands:** import_food_dataset.py imports FMCG packaged food databases into SQLite/Postgres.
- **OpenAPI & Swagger Documentation:** Configured via drf-spectacular at /api/docs/ (Swagger UI) and /api/redoc/.

B. Node.js / Express AI Proxy Server (`server.ts` & `server/`)

- **routes.ts:** High-throughput Express router serving instant barcode lookups from in-memory product collections.
- **geminiVision.ts & geminiNutrition.ts:** Proxies requests to Google Gemini 2.0 Flash and Flash Vision, keeping secret API keys secure on the server side.
- **Binary Magic-Byte Inspection:** Inspects image byte arrays directly (JPEG: FF D8 FF, PNG: 89 50 4E 47, WebP: RIFF...WEBP), rejecting disguised executables or corrupt files.
- **Multer Upload Security:** Enforces an 8MB file size ceiling and validates multipart stream boundaries.

Microservice Endpoint Distribution

Endpoint Route	Port	Runtime	Architectural Responsibility
`/api/analyze/barcode`	8000/5000	Django / Express	Barcode resolution, FSSAI thresholding, personalized medical warnings.
`/api/analyze/plate`	5000	Node / Gemini	Multimodal plate photo segmentation, gram estimation, macro breakdown.
`/api/analyze/ocr`	8000/5000	Django / Node	OCR label text extraction and additive INS identification.
`/api/chat`	5000	Node / Gemini	Conversational clinical nutrition assistant with dietary prompt engineering.
`/api/users/profile`	8000	Django REST	User medical preferences, age, body weight, and scan history sync.

7. Security, Identity & Android Cryptography

FoodAI enforces an end-to-end zero-trust architecture spanning mobile operating system boundaries, cryptographic keystores, network transit, and database tenant isolation.

Security Domain	Technology & Standard	Implementation & Protection Mechanism
Hybrid Google Authentication	Capacitor Google Sign-In & Credential Manager	On native Android, utilizes Credential Manager with OAuth Web Client ID (59556325769-489r0888...) to obtain ID tokens. Web uses Firebase popup. Tokens are verified by Firebase Auth.
Android Keystore Verification	SHA-1 & SHA-256 Signatures in Firebase Console	Debug keystore fingerprints registered in Firebase Console ensure Google Play Services only issues OAuth tokens to authorized APK binaries, preventing repackaging attacks.
Firebase Google Services	google-services.json & Gradle Plugin	Gradle processes google-services.json at build time, generating default_web_client_id, google_app_id, and API keys securely into native resources.
Firestore Database Isolation	Role-Based Security Rules (firestore.rules)	Enforces default-deny. All reads and writes to /users/{uid}, foodScans, and dailyLogs require request.auth.uid == userId and pass schema validation.
Transport Security	TLS 1.3 / HTTPS Encryption	All client-backend and AI proxy traffic is encrypted via TLS 1.3. Android Network Security Configuration prohibits unencrypted plaintext communication outside local debugging.
Offline Fallback Security	Sandboxed Local Demo Session	Provides fully isolated offline operation when network connectivity is lost. Clinical calculation runs locally without exposing credentials or telemetry.

Firestore Security Rule Architecture (`firestore.rules`)

Cloud Firestore security rules prevent data tampering and unauthorized cross-tenant data access:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    // 1. Default-Deny Catch-All
    match /{document=**} { allow read, write: if false; }
    // 2. Security Primitives
    function isSignedIn() { return request.auth != null && request.auth.uid != null; }
    function isOwner(userId) { return isSignedIn() && request.auth.uid == userId; }
    // 3. User Profiles & Subcollections
    match /users/{userId} {
      allow read, write: if isOwner(userId);
      match /foodScans/{scanId} { allow read, write: if isOwner(userId); }
      match /dailyLogs/{logId}   { allow read, write: if isOwner(userId); }
    }
  }
}
```

8. Exhaustive Repository File Map (Frontend & Android)

Complete technical catalog of all Frontend and Android native source files:

Filesystem Path	Module / Component	Engineering Description & Responsibilities
`src/pages/Scan.tsx`	**Vision Scanner Screen**	All-in-one vision camera screen: 60ms auto-detection barcode loop, animated laser, tap-to-focus ring, zoom controls, torch, in-viewfinder OCR snapping, and cart mode.
`src/pages/PlateAnalyzer.tsx`	**Plate AI Meal Vision**	Multimodal plate photography analyzer: calls Gemini Vision, segments food items, computes calories and macros, and provides interactive gram portion sliders.
`src/pages/ProductResults.tsx`	**Product Health Report**	Comprehensive analysis view rendering FSSAI FOPNL badges, personalized medical alerts, ingredient analysis, and tactical damage control mitigation advice.
`src/pages/DailyTracker.tsx`	**24-Hr WHO Tracker**	Tracks daily intake against WHO maximum limits: Free Sugar (25g), Salt (5g), Saturated Fat (20g), and Additives. Features progress rings and weekly charts.
`src/pages/GroceryCart.tsx`	**Basket Screener**	Pre-checkout grocery cart evaluator: calculates Basket Health Grade (A to D), identifies the worst ultra-processed product, and suggests clean-label swaps.
`src/pages/CatalogExplorer.tsx`	**FMCG Catalog Directory**	Search engine across 30+ packaged FMCG brands with filters for Palm-Oil Free, Low Sugar (<5%), Low Salt (<0.3%), and food categories.
`src/pages/ProductComparison.tsx`	**Side-by-Side Compare**	Head-to-head comparison of two packaged food products, highlighting differences in processing tiers, sugar, sodium, and chemical additives.
`src/pages/FoodDNA.tsx`	**Food DNA Visualizer**	NOVA processing spectrum breakdown (Groups 1-4) with an interactive additive risk breakdown.
`src/pages/AINutritionAnalyzer.tsx`	**AI Nutrition Chatbot**	Conversational clinical nutrition assistant powered by Gemini for answering dietary queries, meal planning, and food safety advice.
`src/pages/VoiceLogPage.tsx`	**Voice Meal Logger**	Natural language speech-to-text meal logging page with offline NLP parsing of spoken food items and automatic calorie logging.
`src/pages/ProfilePreferences.tsx`	**Health Preferences**	User medical profile manager: toggles for 8 conditions (Diabetes, BP, CVD, Celiac, etc.), age, weight, and cloud Firestore sync.
`src/pages/History.tsx`	**Scan History Timeline**	Chronological scan history with cloud Firestore synchronization, offline search, and quick re-analysis shortcuts.
`src/pages/Login.tsx & Register.tsx`	**Authentication Pages**	Authentication screens supporting Email/Password, Native Android Google Sign-In with Credential Manager, and Instant Demo Mode.
`src/services/googleAuthService.ts`	**Google Auth Bridge**	Cross-platform Google authentication handler. Uses @capawesome/capacitor-google-sign-in on Android and Firebase popup on Web.
`src/services/firestoreService.ts`	**Firestore Cloud Service**	Cloud Firestore CRUD service for user profiles, food scans, and daily intake logs with robust offline error fallbacks.
`src/services/appUpdateService.ts`	**Release Updater**	Queries GitHub Releases API to detect newer APK versions, displays an in-app update banner, and provides direct download links.
`src/services/voiceFoodParser.ts`	**Voice NLP Parser**	Offline regex and NLP parser extracting food items, portion quantities, and estimated calories from natural speech strings.
`android/app/google-services.json`	**Firebase Android Config**	Official Firebase client configuration mapping com.foodai.app to project food-ai-433b7 and defining OAuth client IDs.
`android/app/src/main/AndroidManifest.xml`	**Android Manifest**	Declares camera, autofocus, and internet permissions, intent filters, cleartext traffic policies, and file providers.

9. Exhaustive Repository File Map (Backend & API Specs)

Complete technical catalog of Backend microservices, configuration, and API endpoint contracts:

Filesystem Path	Module / Role	Engineering Description & Responsibilities
`foodai_backend/food_analysis/health_evaluator.py`	**Clinical Rules Engine**	Evaluates foods against 8 medical conditions, flags contraindications, and generates tactical damage control guidance.
`foodai_backend/food_analysis/nutrition.py`	**FSSAI Math Calculator**	Calculates nutrient weight percentages, FSSAI FOPNL thresholds, and energy contribution ratios.
`foodai_backend/food_analysis/classifier.py`	**Additive Classifier**	Categorizes food additives by INS number and assigns safety toxicity tiers (Clean, Caution, Warning, Danger).
`foodai_backend/firebase_auth.py`	**Firebase Django Auth**	Custom Django authentication backend validating Firebase JWT Bearer tokens and managing user sessions.
`foodai_backend/food_analysis/management/commands/`	**Dataset Importers**	Management commands (import_food_dataset.py) importing FMCG food catalogs into database.
`server/routes.ts`	**Express API Router**	Serves rapid barcode lookups from in-memory product collections and orchestrates AI endpoints.
`server/services/geminiVision.ts`	**Gemini Vision Service**	Interfaces with Google Gemini 2.0 Flash Vision for plate photo segmentation, portion estimation, and macro calculation.
`server/services/geminiNutrition.ts`	**Gemini Nutrition Service**	Powers the AI conversational nutrition assistant with clinical prompt guidelines.
`firestore.rules`	**Firestore Rules**	Production security rules enforcing default-deny, owner isolation (request.auth.uid == userId), and strict schema validation.
`capacitor.config.ts`	**Capacitor Config**	Defines Android package (com.foodai.app), web assets directory, and Google Sign-In plugin scopes.
`FoodAI.apk`	**Android Release Package**	Compiled native Android APK (~9.07 MB) bundled with full native Google Sign-In and modern camera vision.

REST API Endpoints Specification

Method & Route	Backend	Input Request	Response Payload
`POST /api/analyze/barcode`	Django/Express	`{ barcode: string }`	Full ProductAnalysis: FSSAI badges, additives, personalized medical warnings.
`POST /api/analyze/plate`	Node/Gemini	`Multipart form image (≤8MB)`	Segmented food items, gram weights, calories, carbs, protein, fat, and advice.
`POST /api/analyze/ocr`	Django/Node	`{ text: string } or image`	Extracted ingredient tokens, detected INS additives, and health risk ratings.
`GET /api/products`	Django REST	`search, category, brand`	Paginated list of ProductData catalog items.
`GET/PUT /api/users/profile`	Django REST	`Bearer token + profile JSON`	Updated user medical preferences, age, body weight, and scan history.

10. Build, Testing & Strategic Roadmap

Verification & Automated Testing Suite

FoodAI enforces automated testing across all layers of the stack prior to every release build:

- **Frontend Unit & Integration Tests (Vitest):** 91 tests across 19 suites (100% PASS). Validates AuthContext, Google Sign-In service, scanner controls, Plate AI portion sliders, nutrition mathematics, profile autosave, and router transitions.
- **Backend Unit Tests (Pytest):** 45+ Django REST tests (100% PASS). Validates Firebase JWT verification, FSSAI thresholding calculations, condition evaluator logic, and FMCG catalog importers.
- **End-to-End Tests (Playwright):** Automates full user journeys: registration, barcode scanning, cart batch screening, and report generation.

Android APK Compilation & Installation

Stage	Command Line Execution	Outcome & Artifact
1. Web Build	`npm run build (in frontend/)`	Compiles React TypeScript into optimized distribution assets under frontend/dist/.
2. Capacitor Sync	`npx cap sync android`	Copies dist bundle into frontend/android/app/src/main/assets/public/ and updates plugins.
3. Gradle Packaging	`./gradlew assembleDebug`	Assembles native APK package with Google Services resources: FoodAI.apk (~9.07 MB).
4. Device Install	`adb install -r FoodAI.apk`	Installs or updates the application on a connected Android phone with USB debugging.

In-App Updates & Future Strategic Roadmap

In-App Release Updates (`appUpdateService.ts`): Queries the GitHub Releases API on startup. When a newer semantic version is detected, an in-app banner prompts the user to download the updated APK directly.

Strategic Roadmap Deliverables:

• **Continuous Glucose Monitor (CGM) Sync:** Direct Bluetooth Low Energy (BLE) pairing with Abbott FreeStyle Libre and Dexcom G7 to overlay scanned foods directly onto interstitial glucose curves.

• **Wearable Health Integration:** Two-way synchronization with Apple HealthKit and Google Health Connect for dynamic calorie targets based on active metabolic burn.

• **Crowdsourced FMCG Barcode Verification:** Community-driven catalog expansion with algorithmic OCR verification of ingredient lists and nutrition tables.